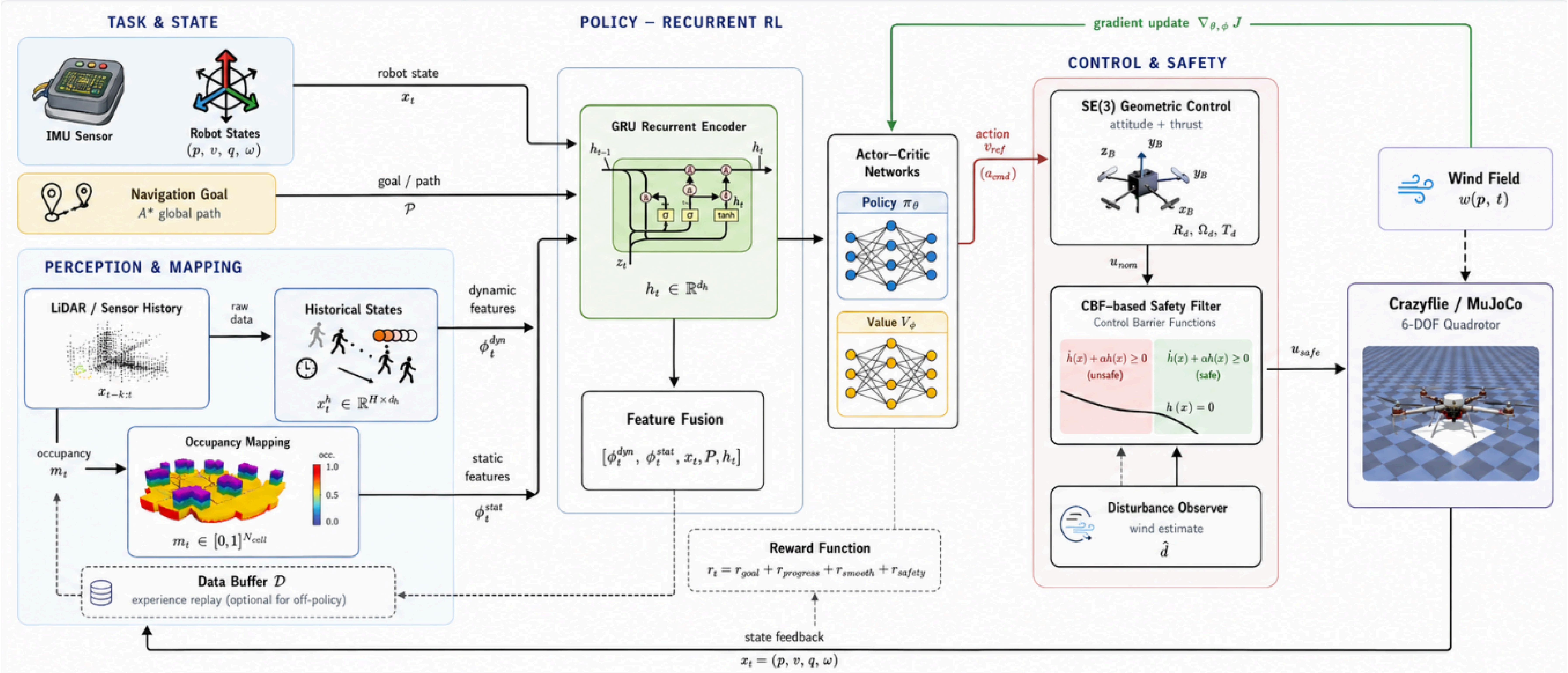

TASK & STATE
IMU Sensor
Robot States
(p, v, q, ω)
Navigation Goal
A^* global path
PERCEPTION & MAPPING
LiDAR / Sensor History
$x_{t-k:t}$
raw data
Historical States
$x_t^h \in \mathbb{R}^{H \times d_h}$
occupancy
m_t
Occupancy Mapping
occ.
1.0
0.5
0.0
$m_t \in [0,1]^{N_{cell}}$
Data Buffer $\mathcal{D}$
experience replay (optional for off-policy)
robot state
x_t
goal / path
$\mathcal{P}$
dynamic features
ϕ_t^{dyn}
static features
ϕ_t^{stat}
POLICY – RECURRENT RL
GRU Recurrent Encoder
h_{t-1}
h_t
z_t
σ
σ
tanh
$h_t \in \mathbb{R}^{d_h}$
Feature Fusion
$[\phi_t^{dyn}, \phi_t^{stat}, x_t, \mathcal{P}, h_t]$
Actor–Critic Networks
Policy π_θ
Value V_ϕ
Reward Function
$r_t = r_{goal} + r_{progress} + r_{smooth} + r_{safety}$
gradient update $\nabla_{\theta,\phi} J$
action
v_{ref}
(a_{cmd})
CONTROL & SAFETY
SE(3) Geometric Control
attitude + thrust
z_B
y_B
x_B
R_d, Ω_d, T_d
u_{nom}
CBF-based Safety Filter
Control Barrier Functions
$\dot{h}(x) + \alpha h(x) \geq 0$
(unsafe)
$\dot{h}(x) + \alpha h(x) \geq 0$
(safe)
$h(x) = 0$
Disturbance Observer
wind estimate
$\hat{d}$
u_{safe}
Wind Field
$w(p, t)$
Crazyflie / MuJoCo
6-DOF Quadrotor
state feedback
$x_t = (p, v, q, \omega)$